DS	Session	Tree 2	Question Information	Level 1	Challenge 71

Question description

Given two rooted trees, your task is to find out if they are *isomorphic*, i.e., it is possible to draw them so that they look the same.

Constraints

- 1≤t≤1000
- 2≤n≤10^5
- the sum of all values of n is at most 10^5

Input

The first input line has an integer t: the number of tests. Then, there are t tests described as follows:
The first line has an integer n: the number of nodes in both trees. The nodes are numbered 1,2,…,n, and node 1 is the root.
Then, there are n−1 lines describing the edges of the first tree, and finally n−1 lines describing the edges of the second tree.

Output

For each test, print "YES", if the trees are isomorphic, and "NO" otherwise.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3
1 2
2 3
1 2
1 3
3
1 2
2 3
1 3
3 2
```

Test Case 2

INPUT (STDIN)

```
2
3
1 2
2 3
1 3
3 3
3
1 2
2 3
1 3
3 3
```

EXPECTED OUTPUT

```
NO
YES
```

EXPECTED OUTPUT

```
NO
NO
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

480

Test Case 3

NLOC

65